





# 1. INTRODUCTION

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For several years now, Social Network Analysis has proved to be a rapidly evolving and expanding field. Only three decades ago, this highly interdisciplinary field was just a narrow chapter of Social Psychology. Today SNA is a commonplace to sociologists, psychologists, anthropologists, economists, and a constantly rising number of people from outside academia.

Along with the popularity of the subject itself, the learning and computational resources related to SNA have grown considerably in both quantity and quality. Still, no matter how sophisticated and powerful they may be, many of the SNA software available worldwide may feel unfriendly and unwelcoming to the non-expert user. Friendliness, simplicity, and an intuitive approach were the main principles I had in mind when I conceived and designed the application described in the present book. Agna is ideal to those who lack experience and a strong theoretical background in social networks.

This software is the product of many years of sweat, curiosity and play. The first versions (1995-2001) were DOS and Windows based and were being used by a very small group of researchers. In the second half of the year 2002 I started to work on version 2. To make this application suitable for any operating system I rewrote it entirely in the Java programming language. Version 2.0 was the first one that I managed to distribute to the community of those interested in Social Network Analysis and Sequence Analysis.

Apart from having a number of bugs fixed, version 2.1 comes with a few improvements and changes, such as the node search utility in the visual network editor. A complete history of these changes can be found on the Agna News page:

• <http://www.geocities.com/imbenta/agna/news.htm>

I really hope that Agna will be helpful to your work, will rise further your desire for knowledge, and will be a cheerful partner in your scientific play. It is my wish that the future versions of this application will have more computing and visualization options and will run faster. As time and resources will be on my side, I will keep working.

I would also like to hear from you and to have your comments on Agna. You are very welcome to e-mail me at [imbenta@yahoo.co.uk](mailto:imbenta@yahoo.co.uk).



## 2. WHAT IS AGNA?

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AGNA (Applied Graph & Network Analysis) is a platform-independent application designed for specific mathematical methods, such as social network analysis, sociometry, and sequence analysis.

Agna can assist you in the study of communication relations in groups, in organization analysis and team building, in the mapping of kinship relations, or in the investigation of the sequential patterns of behaviour.

First, with Agna you can create, edit, store, analyze, and visualize networks. Second, Agna allows you to create networks from behavioral chains.

Agna is a menu-driven application. It consists of two principal frames, each comprising its own set of menus:

- The Main Frame, which is automatically activated when launching the software, contains a spreadsheet and a text editor. The spreadsheet is a square matrix-grid and is located on the left side of the screen. You can use it to enter and edit your network's sociomatrix data. The Output Area is a very simple text editor, which automatically receives analysis output produced by the software.
- The Network Viewer, inactive by default, can be opened by selecting the **View | View Network** option from the Main Frame menu, by typing **Ctrl + Z** or by pressing the ... button. You can use it to visualize and visually edit your networks.

## 3. SETUP

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### 3.1 Prerequisites

Agna can run on virtually any computer with Java Virtual Machine version 1.3 or higher. JVM, also known as J2RE (Java 2 Runtime Environment), is a software that enables your system to run applications written in Java. JVM is available for free on a wide range of platforms, including Windows, Mac OS, Linux, or Solaris. Please note that having a Java-enabled Internet browser doesn't mean that your system supports Java applications.

You can download the JVM from Sun Microsystems:

- <http://java.sun.com/getjava>
- <http://java.sun.com/j2se/1.4/download.html>

The performance of Agna (speed, maximum network size, etc.) strongly depends on your hardware resources. As **minimal requirements**, I suggest a Pentium II processor (or equivalent) with 32 MB RAM and 20 MB free disk space. To test the maximum performance of the software, try creating new networks of various sizes (e.g., 20, 100, 500, 1000 nodes, etc.) using **Ctrl + N**, visualize them by pressing **Ctrl + Z**, and select any command from the **Analysis** menu (e.g., **Sociometrics | Nodal Degree**). When the software is slow, it means you have reached the maximum performance on that system.

### 3.2 Installation

To install Agna on your computer, download the kit from the Agna website:

- <http://www.geocities.com/imbenta/agna>

then save it on your hard disk. MacIntosh users can also download the kit from

- <http://www.tucows.net>

If you encounter problems downloading it, I can send it to you by e-mail

(imbenta@yahoo.co.uk) upon request.

Double-click on the file `Agna_2_kit.jar` – this should open up the Agna Installer window. Enter the path on your computer where you would like Agna to be installed and click **Next**. A new folder called **Agna** will be created automatically and all the files will be extracted there.

If double-clicking on the `Agna_2_kit.jar` file didn't launch the Agna Installer but instead it opened up a browser, a text editor, or an archiver (such as WinZip, PowerArchiver, or Ark), it means that either you don't have the JVM (1.3 or higher) properly installed) or your system implicitly associates the JAR extension with a specific application (text editor, browser, or archiver). Uncompressing the JAR archive will not help. Rather, you may check your JVM and system settings, and try associating the JAR extension with no application. Some archivers on Linux platforms in particular seem to maintain their own lists of file associations aside from the system configuration. So you may want to see the Options settings in your default archiver, too.

If the **Agna** folder already exists (which means, you have an earlier version installed), choose **Update All** and the old files will be replaced by their new versions.

If you don't want or don't like using automated installers, you can extract the files yourself from the ZIP archive `Agna_2_kit.zip`, which can be downloaded from the same locations:

- <http://benta.addr.com/agna/download.htm>
- <http://www.freewebs.com/benta/agna/download.htm>

### 3.3 Running Agna

To launch Agna, double click the file `Agna_2_1.jar`. On Windows platofrms, you can also double-click the executable `Agna_2_1.exe`. Please note that the EXE file still needs a Java Virtual Machine to launch Agna.

If Agna can't be launched using `Agna_2_1.jar` and you are **not** using a Windows operating system, then again, check your JVM and system settings, and try removing any JAR-file association.

### 3.4 Uninstalling Agna

To uninstall this software, simply remove the folder Agna (which was created at installation time) after you have made sure you had no important files you may have stored in any of its subfolders.

## 4. DATA MANAGEMENT

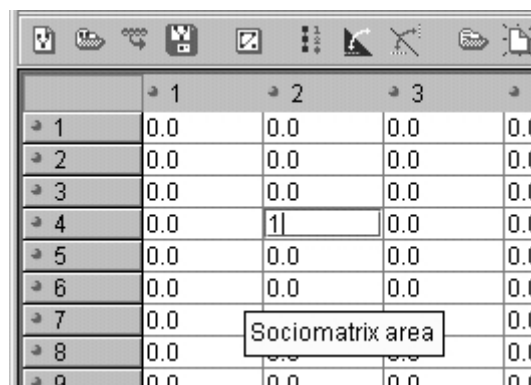
The File menu allows you to create a new network, to open or import an existing network file, to save or export a network you have created, and to extract a network from a behavioural chain file.

Agna treats all sociomatrix data as real numbers. The decision whether a network is binary, weighted, directed, or nondirected, is made by the application at run time. For example, if you wish to compute the indegree coefficients of the current network, Agna first decides whether the network contains weighted or binary data, then computes the suitable coefficients. If the network has weighted data, Agna generates both the simple (binary) and the weighted versions of the indegree coefficient. If the network is based on a zero-one sociomatrix, Agna computes only the simple indegree coefficients.

You have two options to enter and edit your network data: the Main Frame's sociomatrix grid (which is, in fact, a spreadsheet) and the Network Viewer (a visual network editor).

### 4.1 The Spreadsheet

To enter data in the spreadsheet, move the cursor using the arrow keys or by mouse-clicking on a specific grid cell, then type a number. In the figure below, the cell at row 4 and column 2 has just received the value 1, which means a new connection from node 4 to node 2.



|   | 1   | 2   | 3   |     |
|---|-----|-----|-----|-----|
| 1 | 0.0 | 0.0 | 0.0 | 0.0 |
| 2 | 0.0 | 0.0 | 0.0 | 0.0 |
| 3 | 0.0 | 0.0 | 0.0 | 0.0 |
| 4 | 0.0 | 1   | 0.0 | 0.0 |
| 5 | 0.0 | 0.0 | 0.0 | 0.0 |
| 6 | 0.0 | 0.0 | 0.0 | 0.0 |
| 7 | 0.0 |     |     | 0.0 |
| 8 | 0.0 |     |     | 0.0 |
| 9 | 0.0 |     |     | 0.0 |

The diagonal cells of the spreadsheet are not editable, as Agna does not allow any node to be connected to itself. Moreover, Agna always forces the diagonal elements to zero.

You can make selections in the spreadsheet in order to copy, cut, or paste ranges of data. This is the easiest way to let Agna exchange data with other applications.

To edit the name of a node, click on a node label from the vertical header located at the left of the spreadsheet.

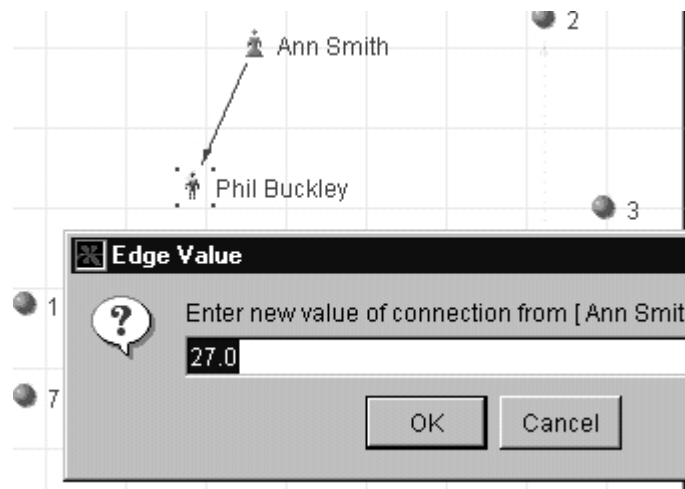
To have all the nodes of the current network renamed as numbers, select **Network | Renumber Nodes** from the menu.

As you will notice, the **Network** menu allows you to apply several transformations on your network, such as: scalar multiplication, square matrix, symmetrization, transposition, and binarization. For more information, please see pages... in this booklet.

## 4.2 The Visual Network Editor

Sometimes you may find it more comfortable to edit your data in the visual mode. Press **Ctrl + Z** to activate the Network Viewer. You can see now the nodes of your network painted on a 400 x 400 pixels area.

To create a connection between two nodes or to set the weight of an already existing edge, click successively on two nodes to have the edge selected (a red arrow will appear), then enter the edge value (eg, by pressing **Ctrl + E**), as in the image below:



In the Network Viewer you can also delete a node using the **Delete** key, clone it by selecting **Data | Clone Node**, and isolate it with **Data | Isolate Node**.

To add a new node to the network, double-clicking anywhere on the image except on an existing node. Nodes can be added also by selecting **Data | Add Nodes**.

To remove all isolated nodes from your network, select **Data | Remove Outsiders** while in the Network Viewer.

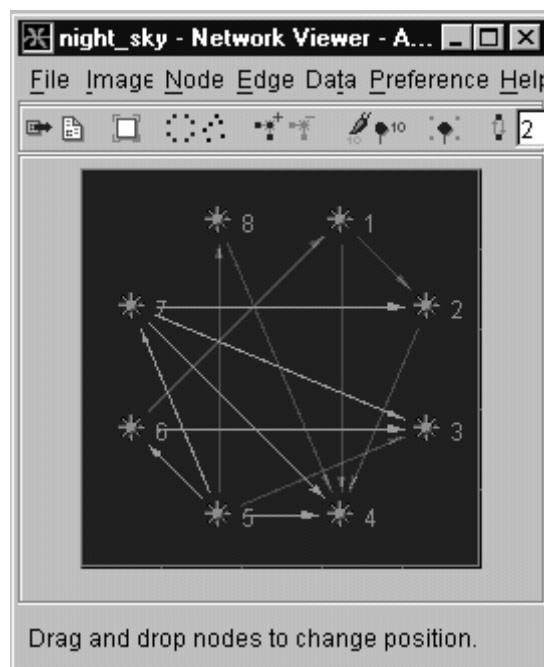
## 5. VISUALIZING NETWORKS

To obtain a visual representation of the current network, simply choose **View | Network Viewer** from the Main Frame menu or press **Ctrl + Z**. This command activates a second frame, which provides you with a simple network visual editor. You can save the network diagram as a file on your hard disk, or insert it into your output file.

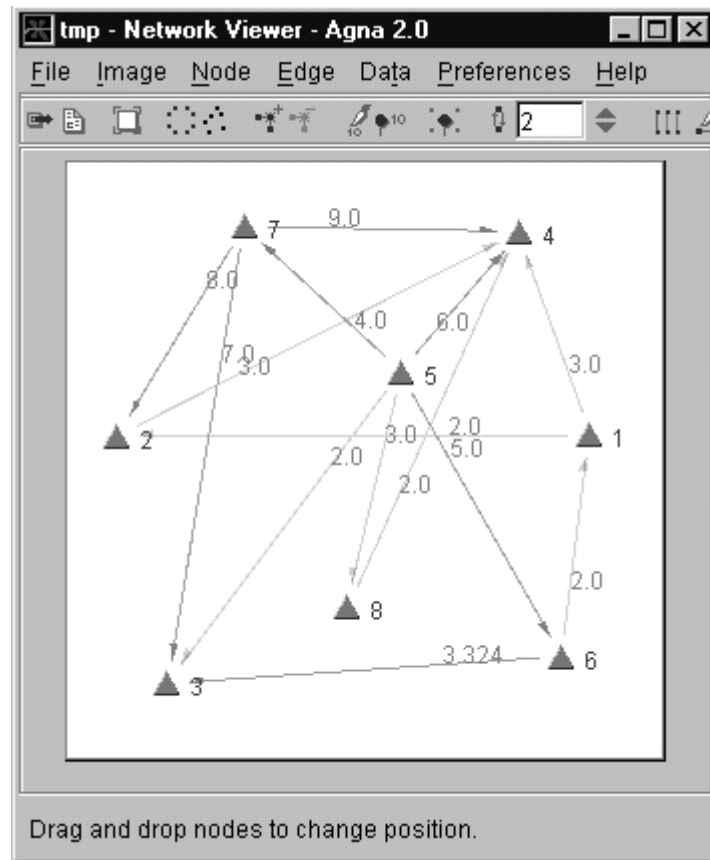
Try to drag and drop a node to change its position. You can move your nodes inside the area to give your network a more intuitive layout.

To customize the image, you can choose from the **Image** menu to: set the image width, scale the image, set the network layer (random vs. circular), change the title attributes of the image, specify the grid lines parameters, the node names, the background settings, the edge loyalty and the face settings for all nodes.

Here you can see the image of an eight-nodes network with green star faces, black background, edge loyalty and circular layout:



And now the same network using different settings: white background, different image width, random layout, blue triangle faces, no loyalty and edge values painted in green:



You can select a node by clicking inside its area (that is, on the node face). Double-clicking on a node opens up a dialogue, which allows you to change the node's position, name, face (ie, graphical appearance), and size. You can also the properties of the selected node using the **Node** menu (p. ...).

By default, clicking successively on two nodes will select an edge. If you don't need this option or you find it annoying, you can disable edge selection (**Edge | Disallow Edge Selection**). The **Edge** menu allows you to set the connection value of the selected edge, the edge separation, the edge colour, the value of the most faded edge, the edge loyalty, and the edge value position and colour.

There are a few operations you can perform in both the sociomatrix editor (Main Frame) and the Network Viewer: adding nodes (double-click on the image outside a node area or by selecting **Data | Add Nodes**), removing outsiders, transposing the network, and symmetrizing it (see pp. ...) . Apart from these commands, the Network Viewer lets you clone, delete, or isolate the currently selected node.

If your network is large, you may find it difficult to identify a specific node or edge. The Search tool located on the left of the image can help you find nodes by name.

To search for a node, type in the search field the name or a fragment of a node name, then hit Enter or press the search button. A list of results will be displayed. You can click on a node from the list and the corresponding node will be selected in the network. A

double-click will show up the node settings dialogue.



## 6. ANALYSING DATA

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All the analytic tools can be accessed through the **Analysis** menu in the Main Frame.

Agna can generate the following computational treatments:

- a primary analysis of the network: a **Basic Description** (p. ...) and some structural parameters: **Diameter** (p. ...), **Density** (p. ...), and **Cohesion** (p. ...);
- various analyses related to the shortest paths in the current network: **Geodesic Matrix** (p. ...), **Shortest Paths...** (p. ...), **All Shortest Paths** (p. ...);
- some node-level coefficients: four types of centrality coefficients (p. ...) and a few sociometric coefficients (p. ...), along with their network-level aggregates and/or statistical descriptions (p. ...).



## 7. FILE MANAGEMENT

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Agna deals with three types of files:

- network data files – ASCII text files used to store and retrieve network data on/from your hard disk, floppy disk, or CD;
- output files – text or HTML files generated by Agna and contain the results of analyses;
- image files – visual representations of the networks generated by the Network Viewer.

### 7.1 Managing Network Data Files

Use the Main Frame's **File** menu to:

- create a new empty network,
- open an existing network from disk,
- create a new network from a chain file, or
- save the current network as a file.

Agna 2.1 supports the following data file formats:

- Agna Data File format (**.agn**)
- Comma Separated Values format (**.csv**)
- Text tab-separated format (**.txt**, **.dat**, **.text**)

#### 7.1.1 Agna Data File Format (**.agn**)

AGN files are ASCII text files that contain full information about the networks you may have created with Agna. AGN files are automatically generated by the application at save time, but you can also edit them manually.

The AGN file format itself does not fall under the Agna general Copyright, which means that other applications are free to export networks in this format.

### 7.1.1.1 A Simple Example

Below is the content of an AGN file in Version 2.1. It represents a 3-node network where every node is connected to each other. The node names are: Max Weber, Emile Durkheim, and Karl Marx. Type the text below in a simple text editor, such as Notepad if you are using Windows, and save it on your hard disk as `Sociologists.agn`.

```
Agna Data File
Version
2.1
This is a fictional network

Network Name
Fathers of Sociology

Network Size
3

Has Viewer
no

Network Matrix
0
1
1
1
0
1
1
1
0
End Network Matrix

Node Names
Max Weber
Emile Durkheim
Karl Marx
End Node Names
```

Don't forget that Notepad may add by default the `.txt` extension to your file name unless you enclose it within quotation marks. Instead of typing the above text, you can also open

with a text editor the file `Sociologists.agn`, which is located in your Agna's `Samples` subfolder.

What you see is the basic structure of an AGN file. It consists of a list of elements starting with the header `Agna Data File`, which is required. The list contains a number of tags – which can be either paired or unpaired – and their respective tag-values. These elements are separated by TAB or LINE BREAK (actually, any combination of characters with ASCII code below 32 would do). For this reason, tags and values cannot contain any TAB or LINE BREAK. Blanc spaces are, however, allowed. For instance, a node can be named Emile Durkheim. Given these rules, you can play with TAB and LINE BREAK to make the file look more readable, as below. Just make sure you never include blanc spaces between tags and tag-values.

```
Agna Data File
This is a fictional network

Version          2.1
Network Name     Fathers of Sociology
Network Size     3
Has Viewer       no

Network Matrix
  0      1      1
  1      0      1
  1      1      0
End Network Matrix

Node Names
  Max Weber
  Emile Durkheim
  Karl Marx
End Node Names
```

Tags can be mentioned in any order you wish. The next element after an unpaired tag must always be its value. Paired tags usually enclose a list of elements, such as the list of node names.

Agna will tacitly ignore any text that is not a tag and is obviously not a tag value (such as `This is a fictional network` in our example).

Please note that tag names are case sensitive!

There is virtually no limit as to the length of a tag-value string.

Try now to open in Agna the file `Sociologists.agn` (`File | Open Network`) and check out the names and the sociomatrix.

## 7.1.1.2 Required tags

Here is the list of the tags that must be present in any AGN file:

|           |                                                                                                                                                                                                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tag name: | <code>Version</code>                                                                                                                                                                                                                                                                                        |
| Pair:     | —                                                                                                                                                                                                                                                                                                           |
| Value:    | <code>Application Version</code> , eg: <code>2.1</code>                                                                                                                                                                                                                                                     |
| Note:     | Useful for backwards compatibility                                                                                                                                                                                                                                                                          |
| Tag name: | <code>Network Name</code>                                                                                                                                                                                                                                                                                   |
| Pair:     | —                                                                                                                                                                                                                                                                                                           |
| Value:    | Name of network, eg: <code>Fathers of Sociology</code>                                                                                                                                                                                                                                                      |
| Note:     | A string                                                                                                                                                                                                                                                                                                    |
| Tag name: | <code>Network Size</code>                                                                                                                                                                                                                                                                                   |
| Pair:     | —                                                                                                                                                                                                                                                                                                           |
| Value:    | Size of network, eg: <code>3</code>                                                                                                                                                                                                                                                                         |
| Note:     | An integer value                                                                                                                                                                                                                                                                                            |
| Tag name: | <code>Has Viewer</code>                                                                                                                                                                                                                                                                                     |
| Pair:     | —                                                                                                                                                                                                                                                                                                           |
| Value:    | <code>yes</code> or <code>no</code>                                                                                                                                                                                                                                                                         |
| Note:     | <code>yes</code> if the file contains graphical information;<br><code>no</code> if the file contains no graphical information                                                                                                                                                                               |
| Tag name: | <code>Network Matrix</code>                                                                                                                                                                                                                                                                                 |
| Pair:     | <code>End Network Matrix</code>                                                                                                                                                                                                                                                                             |
| Value:    | A list of elements separated by TAB<br>or LINE BREAK or combinations of these                                                                                                                                                                                                                               |
| Note:     | These elements represent the network's<br>sociomatrix. Their number must be exactly $g \times g$ ,<br>where $g$ is the value of the <code>Network Size</code> tag.<br>They can be integer, real, positive or negative<br>numbers. Diagonal elements must be 0, or else<br>Agnat will tacitly set them to 0. |
| Tag name: | <code>Node Names</code>                                                                                                                                                                                                                                                                                     |
| Pair:     | <code>End Node Names</code>                                                                                                                                                                                                                                                                                 |
| Value:    | A list of elements separated by TAB or<br>LINE BREAK or combinations of these                                                                                                                                                                                                                               |
| Note:     | These elements must be strings representing<br>the network's node names. Their number must be                                                                                                                                                                                                               |

exactly  $g$ , where  $g$  is the value of the  
`Network Size` tag.

### 7.1.1.3 Using Graphical Information

It is very unlikely that you will manually enter graphical information in AGN files. Rather, you may want to control or check such information without using Agna.

Open now in your text editor the file `SociologistsGr.agn`, which is located in the same `Samples` folder. Basically, this file contains the same network as `Sociologists.agn`, but there is some extra information in it. This time, the `Has Viewer` tag has the value `yes` and the file content is a bit longer:

```
Agna Data File

Version
2.1

Network Name
Fathers of Sociology

Network Size
3

Node Names
Max Weber
Emile Durkheim
Karl Marx
End Node Names

Network Matrix
0
1
1
1
0
1
1
1
0
End Network Matrix
```

Has Viewer  
yes

STG Enabled  
no

Title Visible  
no

Edge Value Visible  
no

Print Names  
yes

Grid Enabled  
no

Face Visible  
yes

Allow Edge Selection  
yes

Color Fidelity  
yes

Area Width  
200

Selected Node  
-1

Second Selected Node  
-1

Names X  
3  
Names Y  
-2

Title X  
13

Title Y  
15

Separator  
0

Edge Value Position  
30

Grid Transparency  
25  
Max Transparency  
35

Background Image X  
0  
Background Image Y  
0  
Background Image Width  
0  
Background Image Height  
0

Grid Space  
40

Edge Value Color  
#00bfa6  
Grid Color  
#0ad20a  
Text Color  
#404040  
Title Color  
#404040  
Background Color  
#ffffff  
Arrow Color  
#89a0a0

Background Image File  
-

Node Faces

```

.\Faces\Light Background\Red Man.gif
.\Faces\Light Background\Green Man.gif
.\Faces\Light Background\Blue Man.gif
End Node Faces

```

```

Node Coordinates
26
30
16
83
90
16
27
153
16
End Node Coordinates

```

### 7.1.1.4 Graphical tags

When graphical information is included in an AGN file, the `Has Viewer` tag must be set to `yes`. The file must provide details about the way Agna should render the network visually. If your AGN file does not provide include information, Agna will use the default rendering preferences.

Here is the list of tags that describe the network visually and set the preferences of the visual network editor's behaviour:

```

Tag name:  STG Enabled
Pair:      —
Value:     yes or no
Note:      Enable/Disable the snap-to-grid function.
           Check out the meanings of yes and no.

```

```

Tag name:  Title Visible
Pair:      —
Value:     yes or no
Note:      Image title's visibility option.
           Check out the meanings of yes and no .

```

```

Tag name:  Edge Value Visible
Pair:      —
Value:     yes or no

```

Note: Edge value's visibility option.  
Check out the meanings of yes and no .

Tag name: **Print Names**  
Pair: —  
Value: **yes or no**  
Note: Node names' visibility option.  
Check out the meanings of yes and no .

Tag name: **Grid Enabled**  
Pair: —  
Value: **yes or no**  
Note: Grid's visibility option.  
Check out the meanings of yes and no .

Tag name: **Face Visible**  
Pair: —  
Value: **yes or no**  
Note: Node faces' visibility option.  
Check out the meanings of yes and no .

Tag name: **Allow Edge Selection**  
Pair: —  
Value: **yes or no**  
Note: Enable/Disable the edge selection function.  
Check out the meanings of yes and no .

Tag name: **Color Fidelity**  
Pair: —  
Value: **yes or no**  
Note: Enable/Disable the color fidelity (or color loyalty function. Check out the meanings of yes and no .

Tag name: **Area Width**  
Pair: —  
Value: Size of image, eg: 200  
Note: An integer value. Meaning .

Tag name: **Selected Node**  
Pair: —  
Value: Index of selected node (end-node), eg: 1  
Note: An integer value from 0 to  $g-1$ . - 1 if none.

Tag name: **Second Selected Node**  
Pair: —  
Value: Index of second selected node (start-node), eg: 1

Note: An integer value from 0 to  $g-1$ . - 1 if none.

Tag name: **Names X**

Pair: —

Value: Horizontal names coordinate (px), eg: 3

Note: An integer value. Meaning .

Tag name: **Names Y**

Pair: —

Value: Vertical names coordinate (px), eg: -2

Note: An integer value. Meaning .

Tag name: **Title X**

Pair: —

Value: Horizontal title coordinate (px), eg: 13

Note: An integer value. Meaning .

Tag name: **Title Y**

Pair: —

Value: Vertical title coordinate (px), eg: 15

Note: An integer value. Meaning .

Tag name: **Separator**

Pair: —

Value: Edge separation, eg: 0

Note: An integer value. Meaning .

Tag name: **Edge Value Position**

Pair: —

Value: Edge value position, eg: 30

Note: An integer between 0 and 100. Meaning .

Tag name: **Grid Transparency**

Pair: —

Value: Grid transparency, eg: 25

Note: An integer between 0 and 255. Meaning .

Tag name: **Max Transparency**

Pair: —

Value: Edge maximum transparency, eg: 35

Note: An integer between 0 and 255. Meaning .

Tag name: **Background Image X**

Pair: —

Value: Background image horizontal coordinate, eg: 0

Note: An integer. Meaning .

Tag name: **Background Image Y**  
Pair: —  
Value: Background image vertical coordinate, eg: 0  
Note: An integer. Meaning .

Tag name: **Background Image Width**  
Pair: —  
Value: Background image width, eg: 0  
Note: An integer. Meaning .

Tag name: **Background Image Height**  
Pair: —  
Value: Background image height, eg: 0  
Note: An integer. Meaning .

Tag name: **Grid Space**  
Pair: —  
Value: Distance between any two grid lines (px), eg: 40  
Note: A positive integer value. Meaning .

Tag name: **Edge Value Color**  
Pair: —  
Value: Edge value color, eg: #00bfa6  
Note: An RGB hex code. Meaning .

Tag name: **Grid Color**  
Pair: —  
Value: Grid color, eg: #0ad20a  
Note: An RGB hex code. Meaning .

Tag name: **Text Color**  
Pair: —  
Value: Node names color, eg: #404040  
Note: An RGB hex code. Meaning .

Tag name: **Title Color**  
Pair: —  
Value: Title color, eg: #404040  
Note: An RGB hex code. Meaning .

Tag name: **Background Color**  
Pair: —  
Value: Background color, eg: #ffffff  
Note: An RGB hex code. Meaning .

Tag name: **Arrow Color**

Pair: —  
 Value: Edge color, eg: #89a0a0  
 Note: An RGB hex code. Meaning .

Tag name: **Background Image File**  
 Pair: —  
 Value: Background Image Path,  
 eg: C:\Picture.jpg (on Windows)  
 Note: Full or relative path. - (a dash) if none. Meaning. Agna 2.x support platform-specific paths only.

Tag name: **Node Faces**  
 Pair: **End Node Faces**  
 Value: A list of image paths separated by TAB  
 or LINE BREAK or combinations of these.  
 Note: Full or relative paths. - (a dash) if none. Their number must be  
 exactly the value of the **Network Size** tag. Meaning. Agna 2.x support  
 platform-specific paths only.

Tag name: **Node Coordinates**  
 Pair: **End Node Coordinates**  
 Value: A list of elements separated by TAB or  
 LINE BREAK or combinations of them.  
 Note: Integers representing horizontal coordinate,  
 vertical coordinate and face width (in this order)  
 for each node. The number of elements must be  
 exactly the value of the **Network Size**  
 tag multiplied by 3.

## 7.1.2 Comma-separated Files (.csv)

A CSV file will contain the sociomatrix and the node names in the form of a matrix of  $g + 1$  LINE-BREAK-separated rows of  $g + 1$  comma-separated values, where  $g$  is the number of nodes. The elements can be placed within quotation marks or not. The first element must be null.

### Example

Open in a text editor the file `Sociologists.csv`, which is located in your `Samples` folder.

```
,Max Weber,Emile Durkheim,Karl Marx
Max Weber,0.0,1.0,1.0
Emile Durkheim,1.0,0.0,1.0
```

Karl Marx,1.0,1.0,0.0

As you can see, it represents a network of three actors whose names are Max Weber, Emile Durkheim and Karl Marx. These actors are all connected among them.

### 5.1.3 Text Tab-separated Files (.txt, .text, .dat)

A text network file will contain the sociomatrix only in the form of a list of  $g \times g$  elements, where  $g$  is the number of nodes. The elements can be separated by TAB, LINE BREAK or any combination of these.

#### Example

Open in a text editor the file `4_full.txt`, which is located in your `Samples` folder.

```
0
2
3
4

5
0
7
8

9
10
0
12

13
14
15
0
```

As you can see, it contains a list of 16 integers, which comprise the sociomatrix of a 4-nodes network, where each node is connected to each other. The values represent the connection weights.

Of course, the same file can be written as a more readable matrix using TABs in the appropriate way:

|   |   |   |   |
|---|---|---|---|
| 0 | 2 | 3 | 4 |
| 5 | 0 | 7 | 8 |

|    |    |    |    |
|----|----|----|----|
| 9  | 10 | 0  | 12 |
| 13 | 14 | 15 | 0  |

## 7.2 Managing Output Files

Once there is some text in the output area, you can save it as an HTML file on disk so that you can view it and print it using an ordinary web browser or load it back into Agna output area.

Agna offers a few editing capabilities for output files: copying, deleting, making selections, and pasting. However, please note that these functions are very limited, as Agna was not designed to work as a real-life Web page editor.

If you have used the Network Viewer's `File | Insert In Output` command, your output contains images. Please note that exporting such an output file as HTML also saves the images as external files. The images are saved in JPEG format in a special folder, which is created in the same folder as the HTML file.

## 7.3 Managing Image Files

You can export any image created by the visual network editor as a SVG or JPEG file on your hard disk, or insert it into the output file (see the Network Viewer's `File` menu).